

PROJECT BRIEF • FORTYGUARD HACKATHON '26

Energy Grid Heat-Demand Forecaster

Predicting tomorrow's power crunch, before it happens.

A plain-language guide to what we're building, why it matters to FortyGuard, how it actually works under the hood, and how our team turns it into a working demo between August 18 and 30.

Team Lead Approved • Moving Forward

What Is It?

In one sentence: software that watches the weather and warns the power grid before it gets overwhelmed by everyone turning on their AC at once.

Think about the hottest day of the year in any city. As the temperature climbs, more and more people switch on air conditioners at roughly the same time. That sudden jump in electricity use is called a **demand spike**. If a spike is big enough and the power company isn't ready for it, the result is a **blackout** — the grid simply can't supply that much power fast enough.

Right now, power companies mostly react to this after it starts happening, or they rely on city-wide weather forecasts that are too broad to say exactly which neighborhoods will spike first. Our project closes that gap.

The Idea in Plain Words

We take FortyGuard's hyperlocal temperature data (accurate down to individual city blocks) and turn it into an early-warning system: "Zone 4 is about to get hot enough that electricity demand there will spike in the next few hours — here's how much, and here's how long you have to prepare."

It's not a weather app. It's a **decision-support agent** — something that watches conditions continuously, figures out when a real risk is forming, and recommends an action to a human, rather than just displaying numbers on a screen.

What Problem Does It Solve?

The aim: give power utilities enough advance warning about heat-driven electricity demand that they can act before things go wrong — not after.

The problem, step by step

- 1. **Heat drives electricity use.** Air conditioning is one of the single biggest drivers of electricity demand in hot climates — and that demand isn't spread evenly across a city.
- 2. **Grids have limits.** Every power grid can only supply so much electricity at once. When demand suddenly outpaces supply, utilities either scramble for emergency power or are forced into rolling blackouts.
- 3. **Current forecasting is too broad.** Most demand planning today uses city-wide or region-wide weather data. It can't tell a utility "this specific district will spike two hours before that one."
- 4. **Reaction is slower than prevention.** By the time a utility sees demand actually rising on their systems, they've already lost the best window to act calmly — rerouting power, alerting large consumers, or bringing backup capacity online.

Why this matters to FortyGuard specifically: Energy Trading is one of only three industries FortyGuard has publicly named as a 2026 growth focus, alongside Agriculture and Rail. This isn't us guessing at relevance — it's a problem they've already said they want solved.

Who this actually helps

Who	What they get
Power utilities / grid operators	Hours of advance warning to pre-position backup capacity or shift load, instead of reacting mid-crisis.
City governments	Fewer blackouts during heatwaves — a direct, visible public-safety win.
Residents & businesses	More reliable power exactly when they need it most: during dangerous heat.
FortyGuard	A working proof-of-concept in a named growth vertical, built by an outside team — strong signal for their roadmap.

How It Actually Works

Five layers, from raw temperature data to a recommendation a human approves.

1 Start with FortyGuard's heat data

We pull current temperature and a 12-hour forecast for each zone we define in a city — this is FortyGuard's core product, already hyperlocal (accurate to roughly city-block level).

2 Apply "Cooling Degree Days" — a real, proven formula

Utilities already use this concept: every degree the temperature rises above a comfortable baseline (commonly around 18–20°C / 65°F), electricity use for cooling goes up in a predictable, well-documented way. This isn't something we're inventing — it's an established method in the energy industry, which makes our model scientifically defensible rather than a guess.

3 Calibrate with public demand data

We layer in publicly available electricity-demand datasets (for example, open grid-operator data) so our model isn't just theoretical — it's tuned against how demand has actually behaved during past heat events.

4 Turn it into a per-zone risk score

For every zone, the system produces a simple 0–100 "Demand Spike Risk" score, updated as new forecast data comes in — along with a plain-language reason (e.g. "Zone 4: temperature rising fast, historically correlates with a 22% demand increase within 3 hours").

5 Wrap it in an autonomous agent loop

The system doesn't just calculate once — it continuously runs a six-step loop: **Observe** conditions → **Evaluate** the risk score → **Plan** a recommended action (e.g. "pre-position backup capacity in Zone 4") → **Validate** against safety guardrails → **Act** by sending the recommendation to a human → **Record** the decision and outcome for a full audit trail. A human always approves before anything is treated as final — the AI recommends, it never acts alone.

Why this counts as an "AI Agent" and not just a chart: The hackathon's official brief calls out AI Agents as a primary track. A dashboard that only displays numbers is a report. Our system continuously watches, reasons, and recommends action — that's the difference between a display and an agent.

PART 4

How We Build It: August 18–30

A realistic day-by-day plan for the two-week sprint.

Dates	Focus	What actually happens
Aug 18–19	Onboard + gather data	Get FortyGuard API access live, pull temperature data for our chosen zones, find and prepare a public electricity-demand dataset to calibrate against.
Aug 20–21	Build the baseline model	Implement the cooling-degree-days formula in code; validate it against historical demand data to confirm it actually predicts real-world patterns.
Aug 22–23	Make it hyperlocal	Turn the baseline into a per-zone, 12-hour-ahead demand-spike risk score, using FortyGuard's forecast data for each zone individually.
Aug 24	Wrap the agent loop	Build the Observe → Evaluate → Plan → Validate → Act → Record cycle, with a human-approval step before any recommendation is treated as final.
Aug 25–26	Build the dashboard	Map view of zones, risk-score cards, a forecast timeline, and the approval interface a "utility operator" would use in the demo.
Aug 27	Guardrails + audit log	Add data-freshness checks, limits on how fast recommendations can change, and full logging of every decision — this is what makes the system trustworthy, not just clever.
Aug 28	Demo scripting	Script a tight 90-second walkthrough; prepare a backup dataset in case live API access is unstable during judging.
Aug 29	Polish + rehearse	Bug fixes, visual clean-up, full team run-throughs of the demo.
Aug 30	Submit + present	Final submission and live pitch to judges.

A rough team split

AI / Modeling Lead

Owens the cooling-degree-days model, demand-data calibration, and the agent's decision logic.

Data & API Integration

FortyGuard API onboarding, zone data pipeline, public demand dataset prep.

Frontend & Dashboard

Map view, zone risk cards, timeline, approval UI.

Agent Logic & Guardrails

The observe-evaluate-plan-validate-act-record loop, thresholds, and audit logging.

Roles can overlap where needed — this is a starting point for discussion, not a fixed assignment.

SUMMARY

Why This Is Worth Our Two Weeks

Question	Answer
What is it?	An AI agent that predicts hyperlocal electricity-demand spikes caused by rising heat, before they happen.
What does it solve?	Blackouts and grid strain during heatwaves, by giving utilities hours of advance warning instead of a reactive scramble.
How does it work?	FortyGuard heat data → a proven "cooling degree days" model → a per-zone risk score → a continuously running agent that recommends action to a human.
Why us?	It's a named FortyGuard growth vertical, it's genuinely hard to fake well, and our team has the AI depth to build the real version instead of a shortcut.

"We're not just showing a chart of temperature — we're giving the grid a head start."